

Advanced Machine Learning

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Exercise 1. Give examples for:

- a) a finite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 2024$. Justify your choice.
- b) an infinite hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) = 2024$. Justify your choice.

Solution:

- a) We know the aspect from Seminar 3 that $\text{VCdim}(\mathcal{H}_{con}^d) = d, \forall d \geq 2$ representing the hypothesis class of boolean conjunctions. In order to find an example where $\text{VCdim}(\mathcal{H}_{con}^d) = d$ we have to set the value d to 2024. The resulting finite hypothesis class $\mathcal{H}$ that has $\text{VCdim}(\mathcal{H}) = 2024$ is $\mathcal{H}_{con}^{2024}$.
- b) From Lecture 6, it is known that the hypothesis class representing the set of all axis-aligned rectangle lying in $\mathbb{R}^d$ is an infinite size class. In addition, it was proven in seminar 3 that $\text{VCdim}(\mathcal{H}_{rec}^d) = 2d$, leading to a choice of d with the value of 1012. Finally, the infinite hypothesis class with $\text{VCdim}(\mathcal{H}) = 2024$ is $\mathcal{H}_{rec}^{1012}$.

Exercise 3. Consider $\mathcal{H} = \mathcal{H}_1 \cup \mathcal{H}_2 \cup \mathcal{H}_3$, where:

$$\mathcal{H}_1 = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid h_a(x) = \mathbf{1}_{[x \geq a]}(x) = \mathbf{1}_{[a, +\infty)}(x), a \in \mathbb{R}\},$$

$$\mathcal{H}_2 = \{h_b : \mathbb{R} \rightarrow \{0, 1\} \mid h_b(x) = \mathbf{1}_{[x < b]}(x) = \mathbf{1}_{(-\infty, b)}(x), b \in \mathbb{R}\},$$

$$\mathcal{H}_3 = \{h_{c,d} : \mathbb{R} \rightarrow \{0, 1\} \mid h_{c,d}(x) = \mathbf{1}_{[c \leq x \leq d]}(x) = \mathbf{1}_{[c,d]}(x), c, d \in \mathbb{R}\}.$$

Consider the realizability assumption. Compute $\text{VCdim}(\mathcal{H})$.

Solution:

Two important aspects were proven in Lecture 6, one of them is that $\mathcal{H}_1$ and $\mathcal{H}_2$ are the threshold functions, with the property $|\mathcal{H}_1| = \infty$ and $|\mathcal{H}_2| = \infty$. The second important aspect is that $\text{VCdim}(\mathcal{H}_1)$ and $\text{VCdim}(\mathcal{H}_2)$ is equal to 1. This is because $\mathcal{H}_1$ and $\mathcal{H}_2$ do not shatter $\mathcal{C}$ as there does not exist a function that can realize the labeling (0,1).

Knowing that $\text{VCdim}(\mathcal{H}_1)$ and $\text{VCdim}(\mathcal{H}_2)$ is 1, the next aspect that we need to look at $\text{VCdim}(\mathcal{H}_3)$. For calculating the $\text{VCdim}(\mathcal{H}_3)$ we need to prove that $\text{VCdim}(\mathcal{H}_3) \geq d$ and $\text{VCdim}(\mathcal{H}_3) < d + 1$.

Firstly, we set $C = \{c_1, c_2\}$ with 2 distinct points, where $c_1 \leq c_2$ with the values $c_1 = 1, c_2 = 3$. In order to obtain the labels (0,0), (0,1), (1,0), (1,1), we can choose different values for c and d from $h_{c,d} \in \mathcal{H}_3$, because the aim is to demonstrate that $\mathcal{H}$ shatters $\mathcal{C}$.

$$\text{For label (0,0), take } c = c_1 - 2, d = c_1 - 1, h_{c,d}(c) = 0, h_{c,d}(d) = 0 \quad (1)$$

$$\text{For label (0,1), take } c = \frac{c_1 + c_2}{2}, d = c_2 + 1, h_{c,d}(c) = 0, h_{c,d}(d) = 1 \quad (2)$$

$$\text{For label (1,0), take } c = c_1 - 1, d = \frac{c_1 + c_2}{2}, h_{c,d}(c) = 1, h_{c,d}(d) = 0 \quad (3)$$

$$\text{For label (1,1), take } c = c_1 - 1, d = c_1 + 1, h_{c,d}(c) = 1, h_{c,d}(d) = 1 \quad (4)$$

From (1), (2), (3) and (4), $\mathcal{H}_3$ shatters $\mathcal{C}$ resulting that $(\mathcal{H}_3) \geq 2$.

Secondly, we set $C = \{c_1, c_2, c_3\}$, with 3 distinct points, with $c_1 \leq c_2 \leq c_3$ and the aim is to find a label that cannot be represented by the hypothesis class. In addition, We have already obtained that the $\text{VCdim}(\mathcal{H}_1) < 3$ and $\text{VCdim}(\mathcal{H}_2) < 3$, meaning that there do not exist any sequences of 3 points that can be shattered by $\mathcal{H}_1$ and $\mathcal{H}_2$. (1)

The label that can be taken into consideration is (1,0,1).

- c_1 has label 1, so: $c_1 \in [c, d]$
- c_2 has label 0, so: $c_2 \notin [c, d]$
- c_3 has label 1, so: $c_3 \in [c, d]$

With this aspect, $\forall c_2 \in [c_1, c_3]$, c_2 is also in $[c, d]$, which leads in assigning c_2 with 1, meaning that here is a contradiction. (2)

Having (1) and (2), means that the $\text{VCdim}(\mathcal{H}) < 3$.

If $\text{VCdim}(\mathcal{H}) < 3$ and we also know that the $(\mathcal{H}_3) \geq 2$, we obtain that $\text{VCdim}(\mathcal{H}) = 2$.

Exercise 4. Consider $\mathcal{H}$ the class of 3-piece classifiers (signed intervals):

$$\mathcal{H} = \{h_{a,b,s} : \mathbb{R} \rightarrow \{-1, 1\} \mid a \leq b, s \in \{-1, 1\}\}, \text{ where } h_{a,b,s}(x) = \begin{cases} s, & x \in [a, b] \\ -s, & x \notin [a, b] \end{cases}$$

- Compute the shattering coefficient $\tau_H(m)$ of the growth function for $m \geq 0$ for hypothesis class $\mathcal{H}$.
- Compare your result with the general upper bound for the growth functions and show that $\tau_H(m)$ obtained at previous point a is not equal with the upper bound.
- Does there exist a hypothesis class $\mathcal{H}$ for which the shattering coefficient $\tau_H(m)$ is equal to the general upper bound (over or another domain $\mathcal{X}$)? If your answer is yes please provide an example, if your answer is no please provide a justification.

Solution:

a.) There are some aspects and indications in Lecture 7 which can lead us on how to solve this exercise. We set $C = \{c_1, c_2, \dots, c_m\}$ and we consider $c_1 < c_2 < \dots < c_m$. In addition, we also know from Lecture 7:

$$\tau_H(m) = \max_{C \subseteq X: |C|=m} |H_C|$$

The parameter s can take the value 1 or -1 , with this in mind we need to count how many cases of labeling we have for each s .

The first case when $s = 1$, we have to count the number of labeling for each sequence of i positive points, where i takes values from 0 to m , that means we assign values for i and show how the sequences of -1 and 1 can appear and to count the labels.

$$\begin{aligned} i = 0 \text{ (no "1" labels)} &: (-1, -1, -1, \dots, -1, -1, -1) \implies 1 \text{ label} \\ i = 1 &: (1, -1, -1, \dots, -1, -1, -1), \dots, (-1, -1, -1, \dots, -1, 1) \implies m \text{ labels} \\ i = 2 &: (1, 1, -1, \dots, -1, -1, -1), \dots, (-1, -1, -1, \dots, 1, 1) \implies (m-1) \text{ labels} \\ &\vdots \\ i = m-1 &: (1, 1, 1, \dots, 1, -1), \dots, (-1, 1, 1, \dots, 1, 1) \implies 2 \text{ labels} \\ i = m &: (1, 1, 1, \dots, 1, 1) \implies 1 \text{ label} \end{aligned}$$

Now we need to sum all the resulting labels and obtain $1 + \frac{m(m+1)}{2}$

For the second case, when $s = -1$ we will take only the labels that do not appear also in the first case, just considering the sequence of i negative points surrounded by positive points, i taking values from 1 to $m-2$.

$$\begin{aligned} i = 1 &: (1, -1, 1, \dots, 1, 1, 1), (1, 1, -1, \dots, 1, 1, 1), \dots, (1, 1, 1, \dots, 1, -1, 1) \implies (m-2) \text{ labels} \\ i = 2 &: (1, -1, -1, \dots, 1, 1, 1), \dots, (1, 1, 1, \dots, -1, 1) \implies (m-3) \text{ labels} \\ &\vdots \\ i = m-2 &: (1, -1, -1, \dots, -1, -1, 1) \implies 1 \text{ label} \end{aligned}$$

Summing the possible labels from the case 2 we obtain $\frac{(m-2)(m-1)}{2}$.

Having the results for both cases we finally compute and get the result

$$\tau_m = 1 + \frac{m(m+1)}{2} + \frac{(m-2)(m-1)}{2} = m^2 - m + 2.$$

b.) $\tau_H(m) = 2^m$ only when $m \leq 3$, so if m is greater than the VC-dimension, then the growth function will increase polynomially, rather than exponentially, according to Sauer's lemma. So, if $m > d+1$, then $\tau_H(m) \leq \sum_{i=0}^d \binom{m}{i}$, this is the general upper bound.

In the Seminar 3, exercise 4, we obtain $\text{VCdim}(\mathcal{H}) = 3$

From the lecture 6, according to the Sauer-Shelah-Perles Lemma, for a hypothesis class $\mathcal{H}$ with $\text{VCdim}(\mathcal{H}) < d$ and all m , we have that:

$$\tau_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i}$$

Having the $\text{VCdim}(\mathcal{H}) = 3$, the general upper bound is:

$$C_m^0 + C_m^1 + C_m^2 + C_m^3 = 1 + m + \frac{m(m-1)}{2} + \frac{m(m-1)(m-2)}{6} = \frac{m^3 + 5m + 6}{6}$$

With the result obtained from a) we now have:

$$m^2 - m + 2 \leq \frac{m^3 + 5m + 6}{6}, \forall m \in \mathbb{N}$$

After calculus we now obtain:

$$(m-1)(m-2)(m-3) \geq 0, \forall m \in \mathbb{N}$$

m	0	1	2	3	
m-1	-	-	0	+	+
m-2	-	-	-	0	+
m-3	-	-	-	-	0
(m-1)(m-2)(m-3)	-	-	0	+	+

Figure 1: Caption

From the sign table, we obtain the solution for the inequality $[1, 2] \cup [3, \infty)$, $\tau_H(m) \leq \sum_{i=0}^3 \binom{m}{i} \forall m \in [1, 2] \cup [3, \infty)$.

c.) There is a hypothesis class H for which the shattering coefficient $\tau_H(m)$ is equal to the general upper bound. This is $H_{thresholds}$, where from Lecture 7, we have obtained $\text{VCdim}(H_{thresholds}) = 1$, the shattering coefficient $\tau_H(m) \leq \sum_{i=0}^1 \binom{m}{i}$. We now have to demonstrate that $\tau_H(m) = \sum_{i=0}^1 \binom{m}{i}$.

Another aspect from Lecture 7 is that $\tau_H(m)$ for $H_{thresholds} = m + 1$.

We have also $\sum_{i=0}^1 \binom{m}{i} = C_m^0 + C_m^1 = m + 1$.

Having all from above in mind, $\tau_H(m) = \sum_{i=0}^1 \binom{m}{i}$, it means that the shattering coefficient is equal to the general upper bound.

Excercise 5. Let $\mathcal{H} = \{h : \mathbb{R} \rightarrow \{0, 1\} \mid h_\theta(x) = \mathbf{1}_{[\theta, \theta+1] \cup [\theta+2, \infty)}(x), \theta \in \mathbb{R}\}$. Compute $\text{VCdim}(\mathcal{H})$.

Solution:

To obtain the $\text{VCdim}(\mathcal{H})$, we have to prove that:

1. There exists a set $\mathcal{C}$ with size d that is shattered by $\mathcal{H}$, where $\text{VCdim}(\mathcal{H}) \geq d$.
2. Every set $\mathcal{C}$ with size $d + 1$ is not shattered by $\mathcal{H}$, where $\text{VCdim}(\mathcal{H}) < d + 1$.

First of all, we can take into consideration the fact that we have 3 possible values for θ . We show that the $\text{VCdim}(\mathcal{H}) \geq 3$ setting $C = \{c_1, c_2, c_3\}$, with the restriction $c_1 \leq c_2 \leq c_3$, proving that $\mathcal{H}$ shatters C .

All the possible labelings of C are the following: $(0,0,0)$, $(0,1,0)$, $(0,0,1)$, $(1,0,0)$, $(1,1,0)$, $(1,0,1)$, $(0,1,1)$, $(1,1,1)$. We have to choose some values for c_1, c_2, c_3 , which are fixed, and different values for θ in order to satisfy all of the possible labels. We consider $c_1 = 0$, $c_2 = 0.9$, $c_3 = 1.5$

For $\theta = 2$ we obtain the label $(0, 0, 0)$:

$$\begin{aligned} c_1 = 0.0 < 2 &\implies h_\theta(c_1) = 0 \\ c_2 = 0.9 < 2 &\implies h_\theta(c_2) = 0 \\ c_3 = 1.5 < 2 &\implies h_\theta(c_3) = 0 \end{aligned}$$

For $\theta = -0.4$ we obtain the label $(1, 0, 0)$:

$$\begin{aligned} c_1 = 0.0 \in [\theta, \theta + 1] &\implies h_\theta(c_1) = 1 \\ c_2 = 0.9 \notin [\theta, \theta + 1] &\implies h_\theta(c_2) = 0 \\ c_3 = 1.5 \notin [\theta, \theta + 1] &\implies h_\theta(c_3) = 0 \end{aligned}$$

For $\theta = 0.2$ we obtain the label $(0, 1, 0)$:

$$\begin{aligned} c_1 = 0.0 < \theta = 0.2 &\implies h_\theta(c_1) = 0 \\ c_2 = 0.9 \in [\theta, \theta + 1] &\implies h_\theta(c_2) = 1 \\ c_3 = 1.5 \notin [\theta, \theta + 1] &\implies h_\theta(c_3) = 0 \end{aligned}$$

For $\theta = 1$ we obtain the label $(0, 0, 1)$:

$$\begin{aligned} c_1 = 0.0 < \theta = 1 &\implies h_\theta(c_1) = 0 \\ c_2 = 0.9 < \theta = 1 &\implies h_\theta(c_2) = 0 \\ c_3 = 1.5 \in [\theta, \theta + 1] &\implies h_\theta(c_3) = 1 \end{aligned}$$

For $\theta = 0.7$ we obtain the label $(0, 1, 1)$:

$$\begin{aligned} c_1 = 0.0 < \theta = 0.7 &\implies h_\theta(c_1) = 0 \\ c_2 = 0.9 \in [\theta, \theta + 1] &\implies h_\theta(c_2) = 1 \\ c_3 = 1.5 \in [\theta, \theta + 1] &\implies h_\theta(c_3) = 1 \end{aligned}$$

For $\theta = 0$ we obtain the label $(1, 1, 0)$:

$$\begin{aligned} c_1 = 0.0 \in [\theta, \theta + 1] &\implies h_\theta(c_1) = 1 \\ c_2 = 0.9 \in [\theta, \theta + 1] &\implies h_\theta(c_2) = 1 \\ c_3 = 1.5 \notin [\theta, \theta + 1] &\implies h_\theta(c_3) = 0 \end{aligned}$$

For $\theta = -0.5$ we obtain the label $(1, 0, 1)$:

$$\begin{aligned} c_1 = 0.0 \in [\theta, \theta + 1] &\implies h_\theta(c_1) = 1 \\ c_2 = 0.9 \notin [\theta, \theta + 1] &\implies h_\theta(c_2) = 0 \\ c_3 = 1.5 \in [\theta + 2, \infty] &\implies h_\theta(c_3) = 1 \end{aligned}$$

For $\theta = -2$ we obtain the label $(1, 1, 1)$:

$$\begin{aligned} c_1 = 0.0 \in [\theta + 2, \infty] &\implies h_\theta(c_1) = 1 \\ c_2 = 0.9 \in [\theta + 2, \infty] &\implies h_\theta(c_2) = 1 \\ c_3 = 1.5 \in [\theta + 2, \infty] &\implies h_\theta(c_3) = 1 \end{aligned}$$

We observe that $\mathcal{C} = \{0, 0.9, 1.5\}$ is shattered by H and we obtain that $\text{VCdim}(\mathcal{H}) \geq 3$.

Second of all, we consider a set $C = \{c_1, c_2, c_3, c_4\}$, with the restriction $c_1 \leq c_2 \leq c_3 \leq c_4$. If $\mathcal{H}$ shatters C , it means that there are 16 possible labels to obtain with different values for θ .

There are labels of 4 elements that can satisfy the conditions, but also, there is one label, $(1, 0, 1, 0)$, which cannot be possible. We try to set $h_\theta(c_1) = 1$, $h_\theta(c_2) = 0$, $h_\theta(c_3) = 1$, $h_\theta(c_4) = 0$ and this would mean that $c_1 \in [\theta, \theta + 1]$, $c_3 \in [\theta + 2, \infty]$ assigned for label 1 and for c_2, c_4 the label 0. It is possible to assign for c_2 label 0 if it is between $[\theta, \theta + 1]$ and $[\theta + 2, \infty]$, but for c_4 , it cannot be assigned label 0 because of the restriction $c_1 \leq c_2 \leq c_3 \leq c_4$ and we know that $c_3 \in [\theta + 2, \infty]$ this means that c_4 would be assigned to label 1, not 0, being a contradiction. This means that:

$$\text{VCdim}(\mathcal{H}) < 4$$

With these aspects in mind, that $\text{VCdim}(\mathcal{H}) \geq 3$ and $\text{VCdim}(\mathcal{H}) < 4$, we find that $\text{VCdim}(\mathcal{H}) = 3$.